

# Multi-Prompt Benchmark Report

16 models x 5 prompt styles (code, iambic pentameter, clinical prose, creative writing, math proof) | NVIDIA Jetson Nano 8GB | llama.cpp 0b1bad1 | GUI off | -ngl 99 -fa on --jinja --temp 0.3 | 2000 token limit

## 1. Generation Speed (tok/s)

Tokens generated per second during inference. Consistent across prompt types because generation is memory-bandwidth bound (reading weights), not compute bound.

| Model           | Params | Quant  | Size     | Code | iambic | Prose | Creative | Math | Avg  |
|-----------------|--------|--------|----------|------|--------|-------|----------|------|------|
| CodeGemma 2B    | 2.51B  | Q4_0   | 1.44 GiB | 30.6 | 30.7   | 30.7  | 30.6     | 30.7 | 30.7 |
| Granite 3.0 2B  | 2.63B  | Q4_K_M | 1.49 GiB | 27.1 | 27.1   | 27.3  | 27.3     | 27.3 | 27.2 |
| Granite 3.2 2B  | 2.53B  | Q4_K_M | 1.44 GiB | 27.2 | 27.3   | 26.9  | 27.2     | 27.2 | 27.2 |
| Gemma 4 E2B     | 4.63B  | Q4_0   | 2.63 GiB | 26.7 | 26.8   | 26.7  | 26.8     | 26.7 | 26.7 |
| Gemma 2 2B      | 2.61B  | Q4_0   | 1.51 GiB | 25.3 | 25.1   | 25.2  | 25.2     | 25.2 | 25.2 |
| LFM 2.5 2.6B    | 2.70B  | Q4_K_M | 1.55 GiB | 25.3 | 25.2   | 25.3  | 25.3     | 25.2 | 25.3 |
| Qwen 2.5 3B     | 3.09B  | Q4_K_M | 1.79 GiB | 21.5 | 21.4   | 21.5  | 21.5     | 21.5 | 21.5 |
| Hermes 3 3B     | 3.21B  | Q4_K_M | 1.87 GiB | 20.8 | 20.8   | 20.8  | 20.8     | 20.8 | 20.8 |
| Llama 3.2 3B    | 3.21B  | Q4_K_M | 1.87 GiB | 20.8 | 20.6   | 20.8  | 20.8     | 20.8 | 20.8 |
| Granite 4 3B    | 3.40B  | Q4_K_M | 1.95 GiB | 19.7 | 19.3   | 19.7  | 19.7     | 19.7 | 19.6 |
| Granite 4.1 3B  | 3.40B  | Q4_K_M | 2.00 GiB | 19.3 | 19.8   | 19.6  | 19.7     | 19.6 | 19.6 |
| Phi-3 3.8B      | 3.82B  | Q4_0   | 2.03 GiB | 20.7 | 21.2   | 20.7  | 20.7     | 20.8 | 20.8 |
| StableLM Zephyr | 2.70B  | Q4_K_M | 1.55 GiB | 27.5 | 28.0   | 27.0  | 28.1     | 26.9 | 27.5 |
| SmallThinker 3B | 3.40B  | Q8_0   | 3.36 GiB | 19.2 | 19.2   | 19.2  | 19.2     | 19.2 | 19.2 |
| Orca-Mini 3B    | 3.02B  | Q4_K_M | 1.90 GiB | 8.0  | 7.5    | 7.3   | 7.6      | 8.6  | 7.8  |
| StarCoder2 3B   | 3.03B  | Q4_K_M | 1.60 GiB | 28.6 | 28.6   | 28.6  | 28.6     | 28.6 | 28.6 |

CodeGemma 2B leads at 30.7 tok/s avg. The 3B class (Qwen, Hermes, Llama, Phi) clusters at 20-22 tok/s. Gemma 4 E2B and Granite 2B bridge the gap at 25-28 tok/s.

## 2. Prompt Eval Speed (tok/s) - Context Ingestion

How fast the model processes the input prompt. Varies by prompt type due to token distribution and attention patterns. Higher = faster time-to-first-token.

| Model          | Code | iambic | Prose | Creative | Math | Avg |
|----------------|------|--------|-------|----------|------|-----|
| CodeGemma 2B   | 438  | 438    | 532   | 589      | 413  | 482 |
| Granite 3.0 2B | 280  | 352    | 298   | 286      | 296  | 302 |
| Granite 3.2 2B | 489  | 462    | 460   | 439      | 495  | 469 |
| Gemma 4 E2B    | 128  | 125    | 141   | 142      | 104  | 128 |
| Gemma 2 2B     | 235  | 236    | 263   | 256      | 218  | 242 |
| LFM 2.5 2.6B   | 233  | 250    | 282   | 282      | 183  | 246 |
| Qwen 2.5 3B    | 380  | 346    | 397   | 421      | 352  | 379 |
| Hermes 3 3B    | 295  | 302    | 410   | 362      | 366  | 347 |
| Llama 3.2 3B   | 378  | 429    | 465   | 474      | 385  | 426 |
| Granite 4 3B   | 307  | 287    | 313   | 298      | 313  | 304 |

| Model           | Code | lambic | Prose | Creative | Math | Avg |
|-----------------|------|--------|-------|----------|------|-----|
| Granite 4.1 3B  | 227  | 275    | 268   | 227      | 246  | 249 |
| Phi-3 3.8B      | 329  | 303    | 415   | 448      | 299  | 359 |
| StableLM Zephyr | 282  | 330    | 368   | 316      | 280  | 315 |
| SmallThinker 3B | 303  | 372    | 316   | 310      | 291  | 318 |
| Orca-Mini 3B    | 118  | 108    | 128   | 122      | 117  | 119 |
| StarCoder2 3B   | 280  | 380    | 344   | 392      | 328  | 345 |

CodeGemma leads at 482 tok/s avg. Gemma 4 E2B is slowest at 128 tok/s (thinking model overhead). Creative/prose prompts generally process faster than code/math across all models.

### 3. Tokens Generated - Output Completeness

Word count of generated output. Lower counts may indicate truncated or incomplete responses. Higher counts suggest more thorough answers (but not necessarily better quality). Re-benchmarked with 2000 token limit and --jinja flag for thinking models.

| Model           | Code | lambic | Prose | Creative | Math | Avg |
|-----------------|------|--------|-------|----------|------|-----|
| CodeGemma 2B    | 185  | 40     | 207   | 283      | 74   | 158 |
| Granite 3.0 2B  | 129  | 133    | 237   | 145      | 250  | 179 |
| Granite 3.2 2B  | 106  | 153    | 691   | 304      | 296  | 310 |
| Gemma 4 E2B     | 256  | 130    | 197   | 221      | 193  | 199 |
| Gemma 2 2B      | 186  | 61     | 268   | 281      | 256  | 210 |
| LFM 2.5 2.6B    | 275  | 168    | 180   | 270      | 200  | 219 |
| Qwen 2.5 3B     | 219  | 54     | 255   | 291      | 274  | 219 |
| Hermes 3 3B     | 175  | 66     | 262   | 278      | 269  | 210 |
| Llama 3.2 3B    | 243  | 47     | 250   | 274      | 238  | 210 |
| Granite 4 3B    | 121  | 39     | 254   | 230      | 249  | 179 |
| Granite 4.1 3B  | 78   | 47     | 282   | 223      | 298  | 186 |
| Phi-3 3.8B      | 141  | 126    | 176   | 235      | 222  | 180 |
| StableLM Zephyr | 166  | 118    | 330   | 138      | 309  | 212 |
| SmallThinker 3B | 845  | 612    | 380   | 779      | 602  | 644 |
| Orca-Mini 3B    | 108  | 64     | 237   | 200      | 78   | 137 |
| StarCoder2 3B   | 1024 | 1005   | 891   | 954      | 146  | 804 |

SmallThinker now produces full output (845 tokens on code) after --jinja fix. Granite models produce clean output after banner-stripping fix. Thinking models (Gemma4, LFM, SmallThinker) spend tokens on internal reasoning.

## 4. Quality Score Matrix (1-10)

Each model's output scored 1-10 by category. Green = 8-10 (excellent), Yellow = 5-7 (adequate), Red = 1-4 (poor/failure). Criteria: Code=correctness+hints+docstring; Iambic=meter+form+imagery; Prose=accuracy+structure+terminology; Creative=sensory+atmosphere+originality; Math=proof+logic+notation.

| Model           | Code | Iambic | Prose | Creative | Math | Avg |
|-----------------|------|--------|-------|----------|------|-----|
| CodeGemma 2B    | 8    | 1      | 2     | 1        | 1    | 2.6 |
| Granite 3.0 2B  | 7    | 4      | 7     | 6        | 8    | 6.4 |
| Granite 3.2 2B  | 8    | 5      | 7     | 7        | 8    | 7.0 |
| Gemma 4 E2B     | 7    | 3      | 6     | 6        | 7    | 5.8 |
| Gemma 2 2B      | 7    | 9      | 8     | 8        | 9    | 8.2 |
| LFM 2.5 2.6B    | 6    | 3      | 7     | 7        | 7    | 6.0 |
| Qwen 2.5 3B     | 8    | 6      | 8     | 8        | 8    | 7.6 |
| Hermes 3 3B     | 7    | 6      | 8     | 8        | 7    | 7.2 |
| Llama 3.2 3B    | 8    | 4      | 8     | 8        | 8    | 7.2 |
| Granite 4 3B    | 7    | 4      | 7     | 7        | 8    | 6.6 |
| Granite 4.1 3B  | 8    | 5      | 7     | 7        | 8    | 7.0 |
| Phi-3 3.8B      | 7    | 5      | 7     | 7        | 8    | 6.8 |
| StableLM Zephyr | 7    | 6      | 7     | 7        | 8    | 7.0 |
| SmallThinker 3B | 7    | 5      | 6     | 6        | 8    | 6.4 |
| Orca-Mini 3B    | 4    | 2      | 5     | 1        | 2    | 2.8 |
| StarCoder2 3B   | 5    | 1      | 1     | 1        | 1    | 1.8 |

Gemma 2 2B is the overall quality leader at 8.2/10 avg, winning or tying for best in 4 of 5 categories. CodeGemma scores 8/10 on code but 1/10 on everything else (chat template loop on non-code prompts). Qwen 2.5 3B is the most consistent 3B model (7.6/10 avg, no score below 6).

## 5. Quality-Speed Metric

Combines quality scores with generation speed. Quality-Speed = (avg quality x avg\_gen\_tps) / 10. Quality-Efficiency = (quality x tokens) / wall\_time. Higher is better. Rewards models that produce high-quality output quickly.

| Model           | Avg Quality (1-10) | Avg Gen (tok/s) | Quality-Speed (Q x tps / 10) | Quality-Efficiency (Q x tok / wall_s) | Rank |
|-----------------|--------------------|-----------------|------------------------------|---------------------------------------|------|
| Gemma 2 2B      | 8.2                | 25.2            | 20.66                        | 90.1                                  | #1   |
| StableLM Zephyr | 7.0                | 27.5            | 19.25                        | 99.9                                  | #2   |
| Granite 3.2 2B  | 7.0                | 27.2            | 19.01                        | 87.6                                  | #3   |
| Granite 3.0 2B  | 6.4                | 27.2            | 17.42                        | 80.9                                  | #4   |
| Qwen 2.5 3B     | 7.6                | 21.5            | 16.32                        | 81.0                                  | #5   |
| Gemma 4 E2B     | 5.8                | 26.7            | 15.51                        | 48.5                                  | #6   |
| LFM 2.5 2.6B    | 6.0                | 25.3            | 15.16                        | 66.5                                  | #7   |
| Hermes 3 3B     | 7.2                | 20.8            | 14.98                        | 70.3                                  | #8   |
| Llama 3.2 3B    | 7.2                | 20.8            | 14.95                        | 71.8                                  | #9   |
| Phi-3 3.8B      | 6.8                | 20.8            | 14.16                        | 55.8                                  | #10  |
| Granite 4.1 3B  | 7.0                | 19.6            | 13.72                        | 64.8                                  | #11  |

| Model           | Avg Quality (1-10) | Avg Gen (tok/s) | Quality-Speed (Q x tps / 10) | Quality-Efficiency (Q x tok / wall_s) | Rank |
|-----------------|--------------------|-----------------|------------------------------|---------------------------------------|------|
| Granite 4 3B    | 6.6                | 19.6            | 12.95                        | 67.9                                  | #12  |
| SmallThinker 3B | 6.4                | 19.2            | 12.29                        | 75.4                                  | #13  |
| CodeGemma 2B    | 2.6                | 30.7            | 7.97                         | 26.0                                  | #14  |
| StarCoder2 3B   | 1.8                | 28.6            | 5.15                         | 22.1                                  | #15  |
| Orca-Mini 3B    | 2.8                | 7.8             | 2.18                         | 12.1                                  | #16  |

Gemma 2 2B wins Quality-Speed at 20.66 - excellent quality at 25.2 tok/s. Granite 3.2 2B is second at 19.01 - best value 2B model. CodeGemma's high speed can't compensate for poor non-code quality (QS=7.97).

## 6. Best vs Worst by Category

Top and bottom performing model for each prompt type, with quality notes explaining the score.

| Prompt Type | Model         | Score | Quality Notes                                                                                                               |
|-------------|---------------|-------|-----------------------------------------------------------------------------------------------------------------------------|
| Code        | CodeGemma 2B  | 8/10  | Correct merge_sort with docstring, type hints, edge cases. Fastest. But generated merge_sort not the requested even-filter. |
|             | Orca-Mini 3B  | 4/10  | Correct logic but terrible formatting (no indentation, no type hints, no docstring). Poor code quality.                     |
| Iambic      | Gemma 2 2B    | 9/10  | Best iambic pentameter. Proper 10-syllable lines, correct iambic stress, 8 lines, ABAB CDCD rhyme. Beautiful imagery.       |
|             | StarCoder2 3B | 1/10  | Base model - echoes prompt and generates generic text completions, not a poem. No iambic pentameter. Not instruction-tuned. |
| Prose       | Gemma 2 2B    | 8/10  | Excellent clinical prose. Well-structured with headers, accurate pathophysiology, good terminology.                         |
|             | StarCoder2 3B | 1/10  | Base model - echoes prompt and generates text completions. No clinical explanation produced. Not instruction-tuned.         |
| Creative    | Gemma 2 2B    | 8/10  | Vivid sensory writing. Strong atmosphere, character detail (Agnes), good pacing. Near-professional quality.                 |
|             | StarCoder2 3B | 1/10  | Base model - echoes prompt. No creative writing produced. Not instruction-tuned.                                            |
| Math        | Gemma 2 2B    | 9/10  | Cleanest proof structure. Theorem header, numbered steps, contradiction clearly shown. Excellent notation.                  |
|             | StarCoder2 3B | 1/10  | Base model - echoes '[proof]' repeatedly. No actual proof. Not instruction-tuned.                                           |

## 7. Summary Recommendations

| Use Case               | Best Model      | Quality     | Gen tok/s | Why                                                                 |
|------------------------|-----------------|-------------|-----------|---------------------------------------------------------------------|
| Best overall quality   | Gemma 2 2B      | 8.2/10      | 25.2      | Highest avg quality, best poetry+math, strong prose+creative        |
| Code generation        | CodeGemma 2B    | 8/10 (code) | 30.7      | Fastest + correct code, but fails on non-code prompts               |
| Best 2B generalist     | Granite 3.2 2B  | 7.0/10      | 27.3      | Consistent quality, fast, good across all categories                |
| All-rounder (3B)       | Qwen 2.5 3B     | 7.6/10      | 21.5      | Consistent 8/10 across prose, creative, math, code                  |
| Creative/prose         | Gemma 2 2B      | 8/10        | 25.2      | Best poetry meter, vivid creative writing, strong clinical prose    |
| Math/proofs            | Gemma 2 2B      | 9/10        | 25.2      | Cleanest proof structure, correct contradiction logic               |
| Reasoning/thinking     | Gemma 4 E2B     | 5.8/10      | 26.7      | Thinking model with reasoning tokens (truncated by 300-token limit) |
| Fastest overall        | CodeGemma 2B    | 2.6/10 avg  | 30.7      | Fastest gen + prompt eval, but code-only quality                    |
| Math reasoning (fixed) | SmallThinker 3B | 8/10 (math) | 19.2      | Full thinking chain then clean formal proof (--jinja + 2000 tok)    |
| Best Quality-Speed     | Gemma 2 2B      | QS=20.66    | 25.2      | Best balance of quality and generation speed                        |
| Best value 2B          | StableLM Zephyr | 7.0/10      | 27.5      | High quality at 2.7B params, good QS=19.25                          |
| Avoid                  | Orca-Mini 3B    | 2.8/10      | 7.5       | Refused creative task, wrong math, poor formatting                  |

| Use Case           | Best Model    | Quality | Gen tok/s | Why                                                    |
|--------------------|---------------|---------|-----------|--------------------------------------------------------|
| Avoid (base model) | StarCoder2 3B | 1.8/10  | 28.6      | Base code model, not instruction-tuned, echoes prompts |

## 8. Output Samples

Representative output previews for quality comparison. Truncated to ~500 chars. Shows best and worst examples per category.

### Iambic Pentameter - Gemma 2 2B (best: 9/10)

The leaves descend, a fiery, vibrant show,  
As summer's warmth gives way to chilling breeze.  
The harvest's bounty, now all brown and low,  
Whispers of winter's icy, silent ease.  
The sun descends, its golden rays now faint,  
And shadows grow, as nature's colors fade.  
The air grows crisp, a frosty, silent paint,  
A blanket of white, upon the land is laid.

### Iambic Pentameter - CodeGemma 2B (failed: 1/10)

user  
assistant  
user  
assistant  
user  
assistant  
user  
assistant  
user  
assistant  
assistant  
user  
assistant  
assistant

### Math Proof - Gemma 2 2B (best: 9/10)

## Theorem:  $\sqrt{2}$  is irrational.  
\*\*Proof by Contradiction:\*\*  
1. **Assumption:** Assume that  $\sqrt{2}$  is rational. - This means we can express  $\sqrt{2}$  as a fraction in its simplest form, where the numerator and denominator are integers with no common factors other than 1. Let's represent this fraction as  $\sqrt{2} = a/b$ , where  $a$  and  $b$  are integers and  $b \neq 0$ .  
2. **Square both sides:** Squaring both sides of the equation gives us:  $2 = a^2/b^2$ .  
3. **Rearrange:** Multiplying both sides by  $b^2$ , we get:  $2b^2 = a^2$ .  
4. \*

### Math Proof - SmallThinker 3B (thinking model, fixed: 8/10)

Alright, I need to prove that the square root of 2 is irrational using a proof by contradiction. I'm a bit rusty on this, but I remember that irrational numbers can't be expressed as a simple fraction, and their decimal representation goes on forever without repeating. So, if I can show that assuming the square root of 2 is rational leads to something impossible, then I've proven it's irrational.  
First, I should start by assuming the opposite, which is that the square root of 2 is rational. If ...

### Code - CodeGemma 2B (specialist: 8/10)

Sure, here's an implementation of the `merge\_sort` function in Python:  
python  
def merge\_sort(lst: list) -> list: """ Sorts a list of integers in ascending order using the merge sort algorithm. Args: lst (list): The list to be sorted. Returns: list: A new list containing the integers in ascending order. Raises: TypeError: If `lst` is not a list. ValueError: If `lst` is empty. Examples: >>> merge\_sort([3, 1, 4, 1, 5, 9, 2, 6])

### Creative - Gemma 2 2B (best: 8/10)

The wind screamed a mournful song against the lighthouse walls, the rain a relentless assault on the glass panes. Inside, Agnes, her face a roadmap of weathered lines, stirred a pot of steaming broth. The scent of salt and seaweed, familiar and comforting, filled the air. She watched the storm rage outside, a solitary figure against the backdrop of the churning sea.  
A heavy thud on the stone steps roused her. A figure, soaked to the bone, stumbled into the light. Agnes's eyes, sharp as flint,